LAYAK

PROJECT 2030 · MYAI HACKATHON · UTM X GCP

LAYAK — The agentic concierge for Malaysian social assistance.

Find out every single subsidy you are eligible for as a Malaysian citizen — in just under two minutes.
One website. Three uploads. Five autonomous steps.
That's all it takes. That's LAYAK.

Track 2 · Citizens First (GovTech & Digital Services)
Version · 24 April 2026

ALIGNED WITH UN SDGS

01 NO POVERTY | 10 REDUCED INEQUALITIES | 16 STRONG INSTITUTIONS

TEAM T010NG

MEMBER 01 HZJ
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Fragmented on the supply side. Invisible to the demand side.

167

social-assistance schemes are currently implemented by 17 ministries and agencies in Malaysia — however, the average citizen is unaware of most of the subsidies available to them. Reason? The effort to search and apply for each one simply isn't worth it.

Source · Ministry of Finance, Economic Outlook 2024

MyGov's Attempt

The MyGov Malaysia super-app's in-house AI chatbot was disabled one day after beta launch (Aug 2025) for hallucinating ministers' portfolios and RON95 prices. Layak avoids this chaos via a strict rule-based AI model. Everything is verified.

Future Facing Project

The Ministry of Digital launched PPDSA (Polisi Pendigitalan Data Sektor Awam) and publicly named *agentic* AI as the next layer for public-sector delivery. Agentic AI is exactly what Layak is built upon.

Meet Aisyah.

Aisyah, 34

GRAB DRIVER · KUANTAN, PAHANG

HOUSEHOLD

Two children (7, 10). One father, 70, resident.

TAX FORM

Form B (self-employed). Deadline 30 June 2026.

SCHEMES IN PLAY

STR 2026 · JKM Warga Emas · 5 LHDN reliefs

INCOME

~RM 2,800 / month gig income. No EA form.

DEVICE

Mid-range Android. Data-capped plan.

TRUST POSTURE

Skeptical after MyGov chatbot incident.

> *"I don't know what I'm entitled to, the forms all want the same documents twice, and I don't trust that entering my IC number anywhere online is safe."*

PRODUCT IN ONE SENTENCE

Endless online searches, replaced with one upload page.

Layak is a grounded agentic concierge: it reads a user's documents, matches them against Budget-2026-gazetted rules, and prepares pre-filled draft packets for all eligible schemes — each one cited to the exact source PDF.

It is explicitly **not** a chatbot. There is no free text answer. Every number comes from a typed rule engine backed by retrieval, and every packet has a *DRAFT — NOT SUBMITTED* disclaimer, to ensure manual user verification before submission.

IN → IC + Payslip + Utility bill
or · Manual entry form

⏱ ~30–60 seconds, 5 visible agent steps
streamed via SSE to the UI

OUT → Ranked schemes · Annual RM upside · 5 draft PDFs
BK-01 · JKM Warga Emas & BKK · LHDN relief summary · STR 2026

FIVE AUTONOMOUS STEPS · ADK-PYTHON SEQUENTIALAGENT

Chat → Action, made visible on stage.

01 · EXTRACT
Read documents.

Pulls name, IC last-4, income, dependants into a typed Profile.

Gemini 2.5 Flash multimodal

02 · CLASSIFY
Tag household.

Income band, children under 18, elderly dependant, form type.

Flash-Lite structured output

03 · MATCH
Check rules.

Typed engine for STR, JKM, LHDN — each rule grounded by search.

Vertex AI Search + rule engine

04 · RANK
Compute RM.

Runs real Python for STR tier, Warga Emas, LHDN tax delta.

Code Execution sandboxed

05 · GENERATE
Draft packet.

Three pre-filled PDFs, watermarked DRAFT — NOT SUBMITTED.

WeasyPrint HTML → PDF

PROVENANCE AS A FIRST-CLASS SURFACE

Every number on screen links to a source PDF.

No rule value ever appears in the UI without a retrievable citation.

This is the single beat that separates Layak from the MyGov Malaysia and NYC MyCity chatbot failures. If Vertex AI Search returns no passage for a rule, that rule is flagged *unverified* and does not contribute to the ranked list.

STR 2026 — Household Tier

PROVENANCE

Household income band	≤ RM 2,500	risalah-str-2026.pdf · p.2
Children multiplier	×2	risalah-str-2026.pdf · p.2
Annual STR cash tier	RM 2,500 / yr	risalah-str-2026.pdf · p.3

retrieved via Vertex AI Search · data store layak-schemes-v1

PDPA 2010-ALIGNED · ZERO-RETENTION BY DEFAULT

Privacy is the primary design constraint.

01 · DATA MINIMISATION

Only eligibility-relevant fields.

Layak reads the fields each rule actually needs — income, dependants, household size — and nothing more. Uploaded documents are parsed in-memory for one request, then discarded. No PII database, no analytics pipeline, no shadow profile.

02 · STATELESS PIPELINE

Documents never leave the request.

Uploaded IC, payslip, and utility-bill files live in memory for the HTTP request only — and are discarded the moment extraction returns.

03 · USER RIGHTS

Export + delete, end-to-end.

Explicit PDPA consent at sign-up, a JSON export endpoint, and a hard-delete endpoint that cascades every evaluation.

FR-17 · FR-12 · FR-18

04 · 30-DAY RETENTION

Free-tier history auto-prunes.

Cloud Scheduler fires a Cloud Run Job every night at 02:00 MYT that deletes free-tier evaluations older than thirty days.

NFR-8 · Nightly prune

05 · NO LIVE SUBMISSION

Draft only. You submit.

Layak never submits on the citizen's behalf. Every output is watermarked *DRAFT — NOT SUBMITTED* on every page.

FR-8 · Disclaimer §7 PRD

06 · OWNER-GATED READS

Firestore rules enforce it.

All writes are backend-only; client reads require `request.auth.uid == resource.data.userId`. A user can only see their own evaluations.

TRD §7 · Security rules

FOR PRIVACY-CONSCIOUS USERS

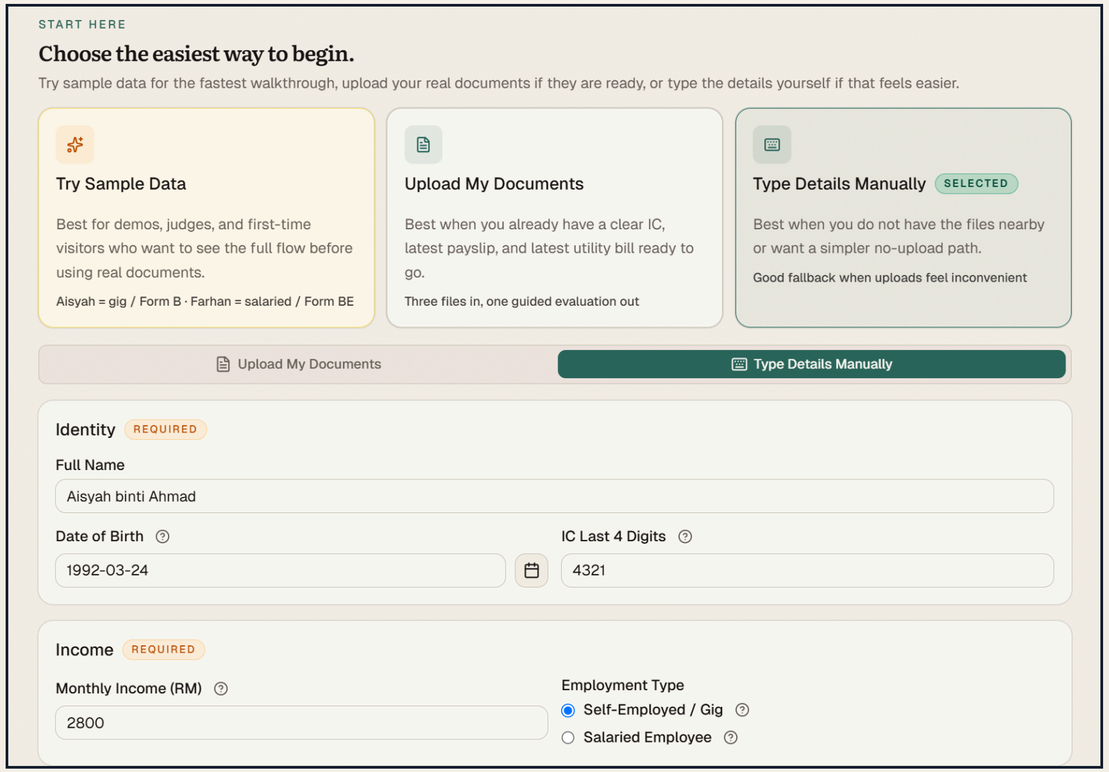
Manual Entry Mode — same pipeline, no upload.

You type it. No MyKad uploads needed.

Worried about handing your documents to an AI? Or don't have the documents conveniently on device? No worries!
A manual input form collects the same info OCR extraction would have produced — then steps 2-5 of the pipeline run unchanged.

We adhere to strict PDPA protocols — no personally identifiable information is stored.

```
Input: Identity | Income | Utility Bills | Dependants
Pipeline: Extract → classify → match → rank → generate
```



HOT-SWAPPABLE MODEL BACKEND

If the data must never leave the country, the model doesn't either.

DEMO BUILD · TODAY

Gemini on Vertex AI

For the hackathon demo we use Gemini 2.5 Flash, Flash-Lite, and Gemini 3 Flash Preview on Vertex AI — we don't have the GPUs to host a strong local model ourselves.

- Multimodal document extraction at 90s target latency
- Gemini Code Execution for visible arithmetic
- Vertex AI Search for grounded retrieval

backend/app/agents/gemini.py · MODEL_ROUTING

PRIVACY-MAXED · ON-PREM

Gemma 4 · local inference

The agentic workflow is model-agnostic by design. For agencies with data-sovereignty requirements, the same ADK-Python pipeline swaps Gemini for **Google's on-device Gemma 4** — no data ever leaves the network boundary.

- Same SequentialAgent, same FunctionTools
- Same typed rule engine, same provenance panel
- Runs entirely inside a trusted ministry perimeter

swap surface · one constant in app/agents/gemini.py

TWO SERVICES ON CLOUD RUN · STATELESS PIPELINE

End-to-end data path.

CLIENT

Browser
Next.js 16 · React 19 · Tailwind 4 · SSE consumer

↓ HTTPS · multipart or JSON ↓

EDGE

Next.js · Cloud Run
Upload widget · SSE proxy · Firebase ID-token attach

FastAPI · Cloud Run
POST /api/agent/intake · min-instances=1 · cpu-boost

↓ ADK SequentialAgent · Gemini 2.5 Pro orchestrator ↓

AGENT TOOLS

extract
Gemini 2.5 Flash · multimodal

classify
Flash-Lite · structured

match
Vertex AI Search + rules

rank
Gemini Code Execution

generate
WeasyPrint → 3 PDFs

↓ grounded retrieval ↓

DATA

Scheme PDFs · git
backend/data/schemes/ · source of truth

Vertex AI Search
layak-schemes-v1 · indexed once

Firestore (v2)
users · evaluations · waitlist

Secret Manager
firebase-admin-key · ADC for Vertex

Eight Google components in one flow.

BRAIN · 01

Gemini 2.5 Pro

RootAgent orchestrator. Holds tools and the conversation buffer.

`ORCHESTRATOR_MODEL`

BRAIN · 02

Gemini 2.5 Flash

Multimodal document extraction. Reads IC, payslip, utility bill directly.

`FAST_MODEL`

BRAIN · 03

Gemini 2.5 Flash-Lite

Household classifier — 5× cheaper than Flash for structured output.

`WORKER_MODEL`

BRAIN · 04

Gemini Code Execution

Sandboxed Python for annual-RM arithmetic — visible in the SSE stream.

`tools: [{codeExecution: {}}]`

CONTEXT · 05

Vertex AI Search

Grounded RAG over the three cached scheme PDFs. Every rule carries a passage.

`layak-schemes-v1 · global`

ORCHESTRATOR · 06

ADK-Python v1.31

First-party GA agent framework. `SequentialAgent` + `FunctionTool`.

`adk deploy cloud_run --with_ui`

LIFECYCLE · 07

Cloud Run

Two services, both min-instances=1 with CPU boost an hour before demo.

`region · asia-southeast1`

IDENTITY · 08

Firebase Auth · v2

Google OAuth only. ID-token verification on every dashboard call.

`firebase_admin.auth.verify_id_token`

OUTCOME · AISYAH'S ANNUAL UPSIDE

RM 12,308

potential relief per year, relative to **RM0** (that she was aware of) prior to using Layak — 5 upside schemes matched + 1 required contribution flagged, computed live by Gemini Code Execution in the demo.

BREAKDOWN · RM / YEAR

JKM Warga Emas Father, 70 · PLI-makanan means test	7,200
JKM BKK Per-child monthly · 2 × RM150	3,600
LHDN Form B 5 reliefs · YA2025 · RM30,500 deduction	558
EPF i-Saraan 15% match · self-employed	500
STR 2026 Household-with-children tier	450
Total upside	**RM 12,308**

PERKESO SKSPS — RM 442.80
REQUIRED CONTRIBUTION · AKTA 789 · PLAN 3

WHAT A GROUNDED PUBLIC-SECTOR AGENT MUST FOLLOW

The rules we built for the agent.

01 **No live submission — ever.**
Layak never writes to any government portal. Outputs are drafts; the citizen submits through the stated official channel.

02 **No unverified claim appears in the UI.**
If Vertex AI Search returns no passage, the rule is flagged *unverified* and does not contribute to the ranked list.

03 **Synthetic demo documents only.**
The MyKad, payslip, and utility-bill specimens are fully fictional and watermarked *SYNTHETIC — FOR DEMO ONLY*.

04 **No final legal determination is claimed.**
Every explanation says "you *appear to* qualify … the agency confirms on application."

05 **AI disclosure is explicit.**
README names Claude Code (Anthropic) as the primary agentic coding assistant, per hackathon Rules §4.2, which is what allowed us to build and ship Layak within a week.

LAYAK · DRAFT APPLICATION PACKET

STR 2026 · Borang BK-01 (Draft)

LHDN (HASiL) · Sumbangan Tunai Rahmah 2026 · https://bantuantunai.hasil.gov.my

Generated: 23 Apr 2026 17:53 UTC
Filer IC (last 4): 4321

Bahagian I — Maklumat Pemohon

Nama Pemohon	AISYAH BINTI AHMAD
Kad Pengenalan (4 akhir)	••••-••-4321
Umur	31 tahun
Jenis Borang Cukai	FORM B
Pendapatan Bulanan	RM 2,800.00

Bahagian II — Isi Rumah

Saiz Isi Rumah	4 orang
Anak Bawah 18 Tahun	2
Tanggungan Warga Emas	Ya
Kategori Pendapatan	B40 HOUSEHOLD WITH CHILDREN

Hubungan	Umur	Kelayakan Anak Bawah 18 (#16a)
Child	10	Ya
Child	7	Ya
Parent	70	—

V1 HACKATHON · V2 SAAS PIVOT · BEYOND

From five schemes to 67. Then 167.

Layak is the reference implementation of the grounded agentic layer PPDSA names.

PPDSA — *Polisi Pendigitalan Data Sektor Awam*, Malaysia's Public Sector Data Digitalisation Policy launched 10 February 2026 — publicly named agentic AI as the next layer above the MyGDX data exchange. Layak is what that layer looks like when it is grounded, auditable, and never submits on the citizen's behalf.

v1	**5 schemes · 2 personas · Form B + BE** STR 2026 · JKM Warga Emas · JKM BKK · LHDN reliefs · PERKESO SKSPS	SHIPPED
v1	**Accounts · history · tiered quotas** Firebase Auth · Firestore · Free 5/24h · Pro unlimited · EN/BM/中文 UI	SHIPPED
v2	**Profile edit · inline PDF preview · mobile** Human-in-loop between extract and classify · structured error recovery	IN BUILD
v3	**Open Weights Model Support & Collaborations** Gemma on-prem swap · ministry perimeter · MyGDX bridge	PENDING
v3+	**Coverage of most schemes** i-Saraan · MyKasih · eKasih · state-level aid · appeal workflow	LONG-TERM

LAYAK

TERIMA KASIH · THANK YOU

Three uploads, one website, *zero hallucinated rules.*

Aisyah gets **RM 12,308** a year she didn't know she was entitled to — from an agent that cites its sources, never submits on her behalf, and runs on Malaysian infrastructure when it has to.

LIVE DEMO

GITHUB

WITH REGARDS,

— *Team T010NG*